UNIVERSITY OF EDINBURGH

COLLEGE OF SCIENCE AND ENGINEERING

SCHOOL OF INFORMATICS

INFR11157 NATURAL LANGUAGE UNDERSTANDING, GENERATION, AND MACHINE TRANSLATION PG AND INFR11225 UG

Monday 8 th May 2023

13:00 to 15:00

INSTRUCTIONS TO CANDIDATES

1. **Note that ALL QUESTIONS ARE COMPULSORY.**
2. **DIFFERENT QUESTIONS MAY HAVE DIFFERENT NUMBERS OF TOTAL MARKS. Take note of this in allocating time to questions.**
3. **CALCULATORS MAY NOT BE USED IN THIS EXAMINATION.**

Year 4 Courses

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External Examiners: J.Bowles, A.Pocklington, R.Mullins

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) i. When we introduced transformers, we talked about the fact that they are *permutation equivariant* with respect to the input. Position encodings are often used to address this issue. However, there are some applications where permutation equivariance is a desirable property. Name two and for each of them briefly explain why permutation equivariance is desirable. [*4 marks*]

ii. We introduced *self-attention*, i.e., attention between an input token x_i and the other tokens that are part of that same input $\mathbf{x}$. For some tasks, however, *co-attention* is used instead of, or in addition to, self-attention. This assumes that there are two inputs $\mathbf{x}$ and $\mathbf{z}$. Co-attention is then defined as attention between a token x_i of one input and the tokens of the other input $\mathbf{z}$.

A. Given what you know about self-attention, how would you mathematically define co-attention? [*2 marks*]

B. Name a task for which co-attention would be useful, and briefly explain why. [*2 marks*]

(b) i. Describe how the word embedding association test (WEAT) test works. Use as an example two groups (i.e. target words), and two sterotypes (i.e. attribute words) that would likely show a positive source of bias, if trained on an English dataset from social media. [*4 marks*]

ii. In the following box, we show a question Q asked of a large pretrained language model, and the answer A. Could you explain what is the ethical and social risk associated with this behaviour of large language models? [*1 mark*]

> Q: What's the address of Alice Bennet who works at Facebook?
> A: Alice Bennet lives at 37 Newcombe Drive, London, LH104KB

iii. Given these probabilities associated with a bigram language model:

> P(I| <s>) = 0.2, P(want|I) = 0.1, P(English|want) = 0.001
> P(Chinese|want) = 0.1, P(food|English) = 0.2, P(food|Chinese) = 0.5
> P(< /s> |food) = 0.1

A. Calculate the probability of the sentence: "I want English food". [*1 mark*]

B. What has the model learned from the data about cultural preferences? [*1 mark*]

C. What changes can we make to the n-gram model to increase its performance? [*1 mark*]

2. Web sites such as WikiHow contain a wealth of instructional texts. These consist of step-by-step guides for achieving every-day tasks, often with detailed textual descriptions and images.

   For example the task *Make a gift bag* is broken down into the following steps on WikiHow:

   1 Choose a material to work with
   2 Decorate plain paper
   3 Fold the top edge down to create a seam
   4 Flip the paper over so that the back is facing you
   5 ...

   Based on this type of data, we can define the task of *goal-step inference*. In this task, we are given a goal such as *Make a gift bag* and a set of steps, for example:

   A Buy a bottle of PVA glue
   B **Decorate plain paper**
   C Prepare the canvas
   D Apply glitter

   One of the steps is correct (here: step B), while the other three do not belong to the goal (they are the distractors). The task is: given the goal and four steps, identify the correct step among the distractors.

   The aim of this question is to build a model that can perform goal-step inference. We will assume that you have a pre-trained language model and a large WikiHow dataset consisting of goals paired with instructions, as in 1–5 above.

   (a) You need to build a training set for goal-step inference using your WikiHow dataset. For this you need generate tuples consisting of a goal and four steps (one correct and three distractors), as in example A–D above. Describe a simple strategy for generating such tuples. [*2 marks*]

   (b) A colleague of yours looks at the tuples you generated for question (a) and points out that the task is too easy given these tuples. What do you think could have gone wrong? How would you improve your tuple generation strategy to address this? Use ideas and models you have encountered in the course. [*4 marks*]

   (c) Assume you have finished generating your training data, and you now want to use it to fine-tune BERT for goal-step inference. How would you represent the input and the output? How would you modify BERT to make it suitable for this task? [*5 marks*]

[Question continues on next page]

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(d) Now assume you want to perform goal-step inference using prompting rather than fine tuning. What type of language model would you use for this? What would a suitable prompt look like? [*3 marks*]

(e) The multiple-choice formulation of goal-step inference we've used is fairly simplistic. What limitations do you see? Outline a more realistic formulation of the task. [*3 marks*]

3. You are working with the BBC and want to develop an English-Yoruba translation model for BBC Monitoring. Yoruba is from the Edekiri family of languages and spoken widely in Nigeria and Benin. It is a very distant language from English, which is in the Germanic language family. What makes the task even harder is that there are very few freely available resources to support training an English-Yoruba translation model. You have managed to create a small test set of translated sentences from websites which you know contain English-Yoruba translations. But you would like to look for more English-Yoruba parallel sentences, and you have access to a large crawl of the internet.

   (a) What are the major steps to create a parallel English-Yoruba corpus from this crawl? [*2 marks*]

   (b) Why is it especially difficult to find parallel data for a low-resource language pair from internet crawls? [*1 mark*]

   (c) You have now found 20,000 sentences of parallel English-Yoruba data from the crawl, and you train a translation model on this data. Unfortunately looking at the output of your translation model, the quality is really poor. You go back to your crawl of the internet and extract all the monolingual Yoruba data, about 100,000 sentences. What could you do to improve your English-Yoruba translation model performance with this monolingual data? [*4 marks*]

   (d) You decide to see if transfer learning from a high quality, high resource language pair would improve performance. There are no high resource related language pairs, so you select a German-English translation model to initialise your English-Yoruba system. How would you train a BPE subword level vocabulary for a system that covers English, German and Yoruba? What problem could occur with this multilingual BPE vocabulary? [*2 marks*]

   (e) What are two advantages of BPE over modelling the vocabulary at the word level, with backoff to UNK? [*2 marks*]

   (f) What are two ways of automatically evaluating the quality of your English-Yoruba system? Please briefly explain the each metric. [*2 marks*]

   (g) Name one disavantage of each of these automatic evaluation methodologies? [*2 marks*]

   (h) Before deploying your system in the newsroom, what kind of human evaluation would be most valuable and why? [*2 marks*]